

4.1 Continuous random variables

Further Probability and Statistics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A probability density integrates to one and probabilities are areas, not heights. For continuous X , $P(X=a)=0$. Find $E(X)=\int xf(x)dx$ and $\text{Var}(X)=E(X^2)-E(X)^2$; transform a distribution by checking domains and Jacobian when needed.

Worked example: follow the reasoning

If $f(x)=2x$ on $0 \leq x \leq 1$, $P(X \leq 0.5) = \int_0^{0.5} 2x \, dx = 0.25$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $f(x)=kx$ on $0 \leq x \leq 2$, zero elsewhere, find k and $P(X>1)$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$1 = \int_0^2 kx \, dx = 2k$, so $k = 1/2$ [2]; $P(X > 1) = \int_1^2 x/2 \, dx = 3/4$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.